

PROJECTS.

1. Blockchain-Based E-Voting System

A secure, transparent, and tamper-proof voting platform built on blockchain technology. Each vote is stored as a transaction on the blockchain, ensuring immutability and verifiability. The system provides real-time vote counting while maintaining voter anonymity.

Tech Stack:

- **Frontend:** React.js, HTML5, CSS3, JavaScript, Web3.js
- **Backend:** Python (FastAPI/Django), Node.js
- **Blockchain:** Ethereum/Solidity (Smart Contracts), Hyperledger
- **Database:** PostgreSQL, IPFS (for distributed storage)
- **Security:** JWT Authentication, Cryptographic Hashing (SHA-256), Digital Signatures
- **Additional:** MetaMask integration, Ganache for testing, Truffle Suite

Features:

- Voter registration with biometric verification
- Real-time vote casting and counting
- Transparent audit trail
- Decentralized result declaration
- Multi-factor authentication

2. E-Commerce Platform

A full-featured online shopping platform with product catalog, shopping cart, payment processing, order management, and admin dashboard. Supports multiple vendors, inventory tracking, and customer reviews.

Tech Stack:

- **Frontend:** React.js, Redux, HTML5, CSS3, Bootstrap/Tailwind
- **Backend:** Python (Django), Node.js (Express)
- **Database:** PostgreSQL, MongoDB, Redis (caching)
- **Payment:** Stripe API, PayPal, M-Pesa API
- **Cloud:** AWS S3 (image storage), CloudFront CDN
- **Additional:** Elasticsearch (product search), RabbitMQ (order queue)

Features:

- Product search and filtering
- Shopping cart and wishlist
- Secure payment gateway
- Order tracking
- Vendor dashboard
- Customer reviews and ratings

3. Weather Forecasting Web App

A location-based weather application providing real-time weather data, hourly and weekly forecasts, severe weather alerts, and interactive weather maps with beautiful animations.

Tech Stack:

- **Frontend:** React.js, JavaScript, HTML5, CSS3, Chart.js
- **APIs:** OpenWeatherMap API, WeatherAPI, Mapbox/Google Maps API
- **Geolocation:** Browser Geolocation API, reverse geocoding
- **State Management:** Redux/Context API
- **Styling:** Styled Components, CSS Modules
- **Additional:** Service Workers (offline support), PWA

Features:

- Current weather conditions
- 7-day forecast
- Interactive radar maps
- Severe weather alerts
- Location search and save favourites
- UV index, air quality, humidity data

4. Notes Saver Application

A note-taking application with rich text editing, categorization, search functionality, and cloud sync. Users can create, edit, delete, and organize notes with tags and folders.

Tech Stack:

- **Frontend:** React.js, JavaScript, HTML5, CSS3, TipTap/Quill (rich text)
- **Backend:** Python (Flask), Node.js

- **Database:** Firebase Firestore, MongoDB
- **Authentication:** Firebase Auth, JWT
- **Storage:** Cloud storage for attachments
- **Additional:** Markdown support, PWA

Features:

- Rich text formatting
- Tags and categories
- Search functionality
- Cloud sync across devices
- Voice-to-text input
- Export to PDF/Word

5. Task Manager Application

A productivity tool for managing tasks and projects with features like task creation, deadlines, priorities, team collaboration, and progress tracking.

Tech Stack:

- **Frontend:** React.js, JavaScript, HTML5, CSS3, Drag-and-drop (react-beautiful-dnd)
- **Backend:** Python (Django REST), Node.js
- **Database:** PostgreSQL, MongoDB
- **Real-time:** WebSockets, [Socket.io](https://socket.io/)
- **Notifications:** Push notifications, Email (Nodemailer)
- **Additional:** Redis for caching, Celery for background tasks

Features:

- Task creation with due dates
- Priority levels (High/Medium/Low)
- Project boards (Kanban style)
- Team collaboration
- File attachments

6. Company Websites

Professional, responsive company websites with corporate information, services, team profiles, contact forms, and blog sections. Designed to establish online presence and generate leads.

Tech Stack:

- **Frontend:** React.js/Next.js, HTML5, CSS3, JavaScript, Bootstrap/Tailwind
- **CMS:** Contentful, Strapi (headless CMS)
- **SEO:** Next.js SSR, meta tags optimization
- **Forms:** Formspree, EmailJS
- **Analytics:** Google Analytics, Hotjar
- **Hosting:** Vercel, Netlify, AWS

Features:

- Responsive design
- Service showcase
- Team profiles
- Blog/News section
- Contact forms
- Google Maps integration

7. Personal Portfolios

Custom-designed portfolio websites for professionals to showcase their work, skills, experience, and achievements. Includes project galleries, resume downloads, and contact forms.

Tech Stack:

- **Frontend:** React.js, HTML5, CSS3, JavaScript, Framer Motion (animations)
- **Styling:** Styled Components, Tailwind CSS
- **Deployment:** GitHub Pages, Vercel
- **Forms:** Formspree, Email integration
- **Additional:** Three.js (3D elements), PDF preview

Features:

- Project showcase
- Skills visualization
- Resume/CV download
- Contact form
- Social media links
- Dark/light theme toggle

8. Pizza Ordering System

An online pizza ordering platform with customizable pizza builder, real-time order tracking, payment integration, and admin dashboard for managing orders and inventory.

Tech Stack:

- **Frontend:** React.js, JavaScript, HTML5, CSS3, Redux
- **Backend:** Python (Django), Node.js
- **Database:** PostgreSQL, MongoDB
- **Real-time:** [Socket.io](https://socket.io/) (order tracking)
- **Payment:** Stripe, M-Pesa API
- **Maps:** Google Maps/Mapbox API (delivery tracking)
- **Additional:** Twilio (SMS notifications)

Features:

- Interactive pizza builder
- Custom toppings selection
- Real-time order tracking
- Delivery address with maps
- Order history
- Admin dashboard

9. Patient Management System

A comprehensive healthcare management system for clinics and hospitals to manage patient records, appointments, prescriptions, billing, and medical history.

Tech Stack:

- **Frontend:** React.js, HTML5, CSS3, JavaScript, Material-UI
- **Backend:** Python (Django/FastAPI), Node.js
- **Database:** PostgreSQL, MongoDB
- **Authentication:** Role-based access (Admin/Doctor/Patient)
- **Security:** HIPAA compliance, data encryption
- **Additional:** PDF generation (reports), Calendar integration

Features:

- Patient registration
- Appointment scheduling
- Electronic Medical Records (EMR)
- Prescription management
- Billing and invoicing
- Lab results integration

10. Machine Learning Model

Custom machine learning models for various applications including prediction, classification, recommendation, and data analysis. Deployed as APIs for integration with web applications.

Tech Stack:

- **Languages:** Python, R
- **Libraries:** TensorFlow, PyTorch, Scikit-learn, Keras
- **Data Processing:** Pandas, NumPy, Matplotlib
- **Deployment:** Flask/FastAPI, Docker
- **Cloud:** AWS SageMaker, Google AI Platform
- **Additional:** Jupyter Notebooks, MLflow

Features:

- Data preprocessing and cleaning
- Model training and validation
- API endpoints for predictions
- Model versioning
- Performance monitoring
- Automated retraining

11. SACCO System (Savings and Credit Cooperative)

A financial management system for Savings and Credit Cooperative Organizations (SACCOs) to manage member accounts, savings, loans, dividends, and financial reporting.

Tech Stack:

- **Frontend:** React.js, HTML5, CSS3, JavaScript, Redux
- **Backend:** Python (Django), Node.js
- **Database:** PostgreSQL, MySQL
- **Security:** End-to-end encryption, SSL
- **Reports:** JasperReports, PDF generation
- **Integration:** M-Pesa API, Bank APIs
- **Additional:** Excel/CSV export, Audit logs

Features:

- Member registration and management
- Savings accounts and deposits
- Loan application and approval
- Interest calculation
- Dividend processing
- Financial reporting
- Mobile money integration
- Audit trail and compliance

Summary of Technologies Used:

Technology	Projects Used In
React.js	All projects
Python	Blockchain, E-commerce, Task Manager, Patient System, ML Model, SACCO
JavaScript	All projects
HTML5/CSS3	All projects
Node.js	E-commerce, Notes, Task Manager, Pizza System
PostgreSQL	Blockchain, E-commerce, Task Manager, Patient System, SACCO
MongoDB	E-commerce, Notes, Pizza System
Django	E-commerce, Patient System, SACCO
FastAPI	Blockchain, ML Model
Redux	E-commerce, Pizza System, SACCO
Map API	Weather App, Pizza System
Stripe/PayPal	E-commerce, Pizza System
M-Pesa API	E-commerce, Pizza System, SACCO
Docker	ML Model, Backend services
AWS	E-commerce, ML Model
TensorFlow/PyTorch	ML Model
<u>Socket.io</u>	Task Manager, Pizza System